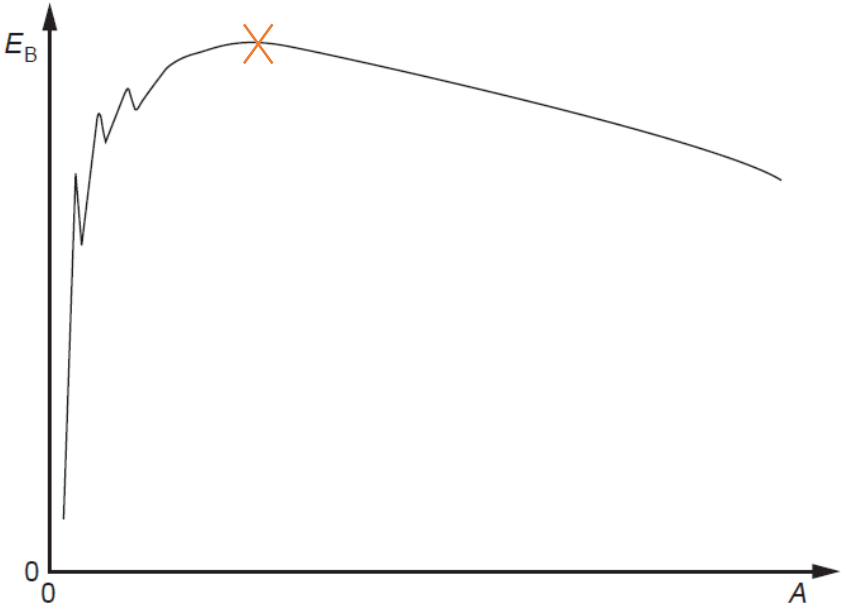


8	(a)	Binding energy = [Total mass of the unbounded nucleons - mass of Iron nucleus]c^2 $BE = [(26 \times 1.00728 + 30 \times 1.00866) - 55.92132]uc^2$ $= (0.52776)(1.66 \times 10^{-27})(3 \times 10^8)^2$ $= 7.88 \times 10^{-11} \text{ J}$ $BE \text{ per nucleon} = \frac{7.88 \times 10^{-11}}{56} = 1.4 \times 10^{-12} \text{ J} = \frac{1.4 \times 10^{-12}}{1.6 \times 10^{-19}} \text{ eV}$ $= 8.8 \text{ MeV}$	B1 M1 M1 A0
	(b)	(i) Near to the highest point on the graph 	B1
		(ii)1. Neutron	B1
		(ii)2. Since the nucleus are of nucleon numbers <u>less than Fe-56</u>, base on the graph, the <u>binding energy per nucleon increases with higher nucleon number</u>. <u>The energy used to break the hydrogen isotopes are thus less than the energy released use to form helium nucleus</u>	B1 B1

	(iii)	Energy released = BE of products – BE of reactants $= (\text{mass defect of products} - \text{mass defect of reactants})c^2$ $= (0.030377 - 0.002388 - 0.009105)uc^2$ $= 2.8 \times 10^{-12} \text{ J}$	C1 A1
(c)		<u>If there is no other particle, beta particle would always have the same energy.</u> <u>By conservation of energy and momentum,</u> <u>there must be another particle to share energy/momentum</u>	B1 M1 A1
(d)	(i)	Comparing the interval between $N = 12 \times 10^{12}$ and $N = 12 \times 10^{12}$, $\Delta t = 0.52 \text{ s}$ Therefore, half-life = 0.52 s $\lambda = \frac{\ln 2}{t_{1/2}} = \frac{\ln 2}{0.52}$ $= 1.3 \text{ s}^{-1}$	M1 A1
	(ii)	Check that number of thorium + protactium is always N (marker to check they intersect at 12, and two more values to add to 24×10^{12}) Increasing graph with decreasing gradient	B1 B1
	(iii)	initial activity = $(\lambda)(\text{initial number of nucleons})$ $= (1.3)(24 \times 10^{12})(= 3.1 \times 10^{13})$ Activity at 1.5 s = $3.1 \times 10^{13} e^{-(1.3)(1.5)}$ $= 4.4 \times 10^{12} \text{ Bq}$	C1 C1 A1
	(iv)	The Geiger counter was only used to count the radioactive particle that enters the GM tube. <u>Since the radioactive particle are released in random direction, not all the radioactive particle are captured by the counter.</u>	B1